OBJECT ORIENTED PROGRAMMING [CT-260]

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DEPARTMENT: BCIT

BATCH: 2025

YEAR & SECTION: FSCS-D

LAB: 3

CODES

Q1.

```
#include <iostream>
#include <cstring>
using namespace std;

class Employee {
private:
    char* EmployeeName;
    const int EmployeeId;   // Cannot be changed after
initialization

public:
    // Constructor with member initializer list
    Employee(const char* name, int id) : EmployeeId(id) {
        EmployeeName = new char[strlen(name) + 1];
        strcpy(EmployeeName, name);
    }

    // Destructor
    ~Employee() {
        delete[] EmployeeName;
    }

    // Getter for name
    const char* getEmployeeName() const {
        return EmployeeName;
    }

    // Getter for ID
```

```
    int getEmployeeId() const {
        return EmployeeId;
    }

    // Setter for name only
    void setEmployeeName(const char* name) {
        delete[] EmployeeName;
        EmployeeName = new char[strlen(name) + 1];
        strcpy(EmployeeName, name);
    }
};

int main() {
    Employee Employee1("Ali", 101);
    Employee Employee2("Sara", 102);
    Employee Employee3("Jhon", 103);

    Employee1.setEmployeeName("Ahmed");

    cout << "Employee 1: " <<
Employee1.getEmployeeName()
         << ", ID: " << Employee1.getEmployeeId() << endl;

    cout << "Employee 2: " <<
Employee2.getEmployeeName()
         << ", ID: " << Employee2.getEmployeeId() << endl;

    cout << "Employee 3: " <<
Employee3.getEmployeeName()
         << ", ID: " << Employee3.getEmployeeId() << endl;

    return 0;
}
```

Q2.

```
#include <iostream>
using namespace std;

class DynamicArray {
private:
    int* arr;
    int capacity;
    int currentSize;

public:
    DynamicArray(int size) {
        capacity = size;
        currentSize = 0;
        arr = new int[capacity];

        for(int i = 0; i < capacity; i++)
            arr[i] = 0;
    }

    ~DynamicArray() {
        delete[] arr;
    }

    void push(int value) {
        if(currentSize < capacity) {
            arr[currentSize++] = value;
        } else {
            cout << "Array is full!" << endl;
        }
```

```
    }

    int size() const {
        return currentSize;
    }

    void display() const {
        for(int i = 0; i < currentSize; i++)
            cout << arr[i] << " ";
        cout << endl;
    }
};

int main() {
    DynamicArray obj(5);
    obj.push(10);
    obj.push(20);
    obj.push(30);

    obj.display();
    cout << "Size: " << obj.size() << endl;

    return 0;
}
```

Q3.

```
#include <iostream>
using namespace std;

class Account {
private:
    int account_no;
    double account_bal;
    int security_code;

    static int count;

public:
    void initialize(int no, double bal, int code) {
        account_no = no;
        account_bal = bal;
        security_code = code;
        count++;
    }

    void display() const {
        cout << "Account No: " << account_no << endl;
        cout << "Balance: " << account_bal << endl;
        cout << "Security Code: " << security_code << endl;
    }

    static int getCount() {
        return count;
    }
};
```

```
int Account::count = 0;

int main() {
    Account a1, a2;

    a1.initialize(1001, 5000.75, 1234);
    a2.initialize(1002, 7000.50, 5678);

    a1.display();
    a2.display();

    cout << "Total Accounts Created: " <<
Account::getCount() << endl;

    return 0;
}
```

Q4.

```
#include <iostream>
using namespace std;

class Student {
private:
    string name;
    const int rollNumber;   // Cannot change
    float gpa;

public:
```

```
    Student(string n, int r, float g) : rollNumber(r) {
        name = n;
        gpa = g;
    }

    void setGPA(float g) {
        gpa = g;
    }

    float getGPA() const {   // const member function
        return gpa;
    }

    int getRollNumber() const {
        return rollNumber;
    }

    void display() const {
        cout << "Name: " << name << endl;
        cout << "Roll Number: " << rollNumber << endl;
        cout << "GPA: " << gpa << endl;
    }
};

int main() {
    Student s1("Ali", 101, 3.5);
    s1.setGPA(3.8);
    s1.display();
    return 0;
}
```

Q5.

```
#include <iostream>
using namespace std;

class HotelMercato {
private:
   string customerName;
   int days;
   static const double rentPerDay;   // Fixed rent decided by
committee
   double totalRent;

   void calculateRent() {
      if(days > 7)
         totalRent = (days - 1) * rentPerDay;
      else
         totalRent = days * rentPerDay;
   }

public:
   HotelMercato(string name, int d) {
      customerName = name;
      days = d;
      calculateRent();
   }

   void display() const {   // Cannot modify data
      cout << "Customer Name: " << customerName << endl;
      cout << "Days: " << days << endl;
      cout << "Rent: " << totalRent << endl;
```

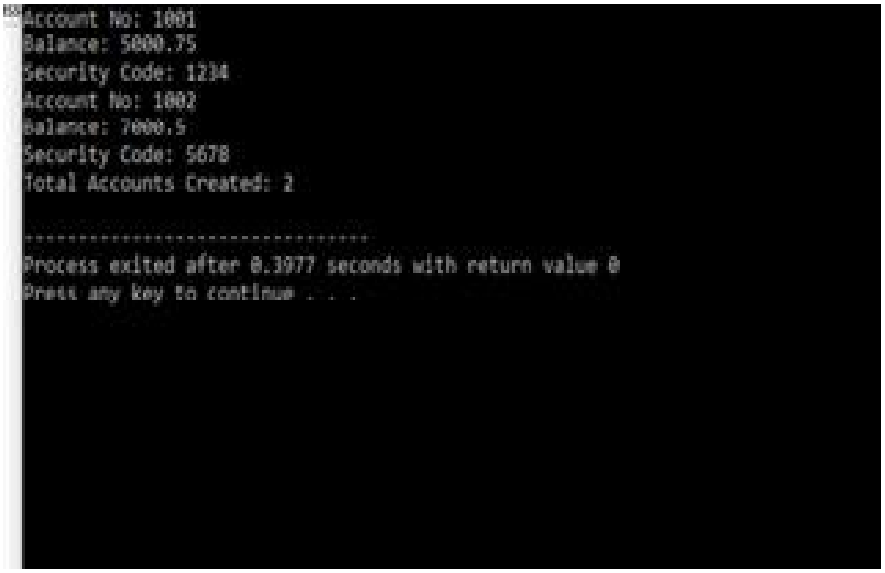
```
    }
};

const double HotelMercato::rentPerDay = 1000.85;

int main() {
    HotelMercato c1("Ahmed", 5);
    HotelMercato c2("Sara", 10);

    c1.display();
    cout << endl;
    c2.display();

    return 0;
}
```

OUTPUTS

Q1.

Q2.

Q.3


Account No: 1001
Balance: 5000.75
Security Code: 1234
Account No: 1002
Balance: 7000.5
Security Code: 5678
Total Accounts Created: 2
Process exited after 0.3977 seconds with return value 0
Press any key to continue . . .

Q4.


Name: Ali
Roll Number: 101
GPA: 3.8
Process exited after 0.1801 seconds with return value 0
Press any key to continue . . .

Q5.


Customer Name: Ahmed
Days: 5
Rent: 5004.25
Customer Name: Sara
Days: 10
Rent: 9007.65
Process exited after 0.7360 seconds with return value 0
Press any key to continue . . .
QUESTION 5.cpp